Technical Appendices for Parameterized Abstract Interpretation for Transformer Verification

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1 Proofs

Due to the length of the formulas in the proof, the appendix is presented in a single-column format.

1.1 Proof of Theorem 1

Our proof proceeds by the following steps: 1) A change of variables, under which the gap G becomes easier to evaluate; 2) A reduction from an infinite number of non-negativity constraints (one for each (x, y) in $[l_x, u_x] \times [l_y, u_y]$) to a single constraint, followed by elimination of that constraint; 3) Solution of the simplified problem that results; 4) Transformation back to the original variables. Along the way, some more involved calculations are captured in three lemmas and their proofs.

1.1.1 Change of variables

We begin by making a change of variables that will simplify evaluation of the mean and minimum gap. We express (x, y) as

$$x = \epsilon_x x' + m_x, \qquad y = \epsilon_y y' + m_y, \tag{1}$$

where now $x', y' \in [-1, 1]$. This changes the function to be bounded to

$$xy = (\epsilon_x x' + m_x)(\epsilon_y y' + m_y) = \epsilon_x \epsilon_y x' y' + m_y \epsilon_x x' + m_x \epsilon_y + m_x m_y \coloneqq F(x', y'). \tag{2}$$

Since (1) is a linear transformation, the lower bound L remains a quadratic and convex [1, Sec 3.2.2] function of (x', y'), which we parameterize in the same way as in (9) in the main text:

$$L(x', y') = \frac{1}{2} \begin{bmatrix} x' & y' \end{bmatrix} \begin{bmatrix} A'_{xx} & A'_{xy} \\ A'_{xy} & A'_{yy} \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} b'_x & b'_y \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + c', \qquad A' \succeq 0, \tag{3}$$

with parameters $A' \succeq 0$, b', c'. The gap $G(x', y') = F(x', y') - L(x', y')$ is then

$$G(x', y') = \frac{1}{2} \begin{bmatrix} x' & y' \end{bmatrix} \begin{bmatrix} -A'_{xx} & \epsilon_x \epsilon_y - A'_{xy} \\ \epsilon_x \epsilon_y - A'_{xy} & -A'_{yy} \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} \underbrace{m_y \epsilon_x - b'_x}_{-\tilde{b}_x} & \underbrace{m_x \epsilon_y - b'_y}_{-\tilde{b}_y} \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \underbrace{m_x m_y - c'}_{-\tilde{c}}. \tag{4}$$

For computing the mean gap $\mathbb{E}[G(x', y')]$, the linearity of (1) also implies that the distribution of (x', y') remains uniform, namely $(x', y') \sim U([-1, 1]^2)$. This results in the following simpler expression for $\mathbb{E}[G(x', y')]$.

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Lemma 1. *Under the change of variables to (x', y') and the parameterization in* (4)*, the mean gap is given by*

$$\mathbb{E}[G(x', y')] = -\frac{1}{6}(A'_{xx} + A'_{yy}) - \tilde{c}. \tag{5}$$

Proof. Since $(x', y') \sim U([-1, 1]^2)$, x' and y' are zero-mean and statistically independent, hence $\mathbb{E}[x'y'] = 0$ also. It follows that in taking the expectation of (4), the only non-zero terms are

$$\mathbb{E}\big[x'^2\big] = \mathbb{E}\big[y'^2\big] = \frac{1}{3},$$

as well as the constant term $-\tilde{c}$. Combining this with the coefficients in (4) yields (5). □

1.1.2 Reduction of constraints

Reduction from infinite constraints to a single constraint The non-negativity constraint, now $G(x', y') \geq 0$ for all $(x', y') \in [-1, 1]^2$, is equivalent to

$$\min_{(x',y')\in[-1,1]^2} G(x', y') \geq 0. \tag{6}$$

We show that the left-hand side simplifies as follows.

Lemma 2. *Under the parameterization of $G(x', y')$ in* (4)*,*

$$\min_{(x',y')\in[-1,1]^2} G(x', y') = -\frac{1}{2}(A'_{xx} + A'_{yy}) + \min\left\{\epsilon_x\epsilon_y - A'_{xy} - \left|\tilde{b}_x + \tilde{b}_y\right|, A'_{xy} - \epsilon_x\epsilon_y - \left|\tilde{b}_x - \tilde{b}_y\right|\right\} - \tilde{c}. \tag{7}$$

Proof. The lemma results from showing that the minimum in (6) occurs at one of the four vertices $(x', y') = (\pm 1, \pm 1)$. To show this, we consider the Hessian of $G(x', y')$, which is the matrix

$$\nabla^2 G(x', y') = \begin{bmatrix} -A'_{xx} & \epsilon_x\epsilon_y - A'_{xy} \\ \epsilon_x\epsilon_y - A'_{xy} & -A'_{yy} \end{bmatrix}.$$

This matrix cannot be positive semidefinite, as that would require $-A'_{xx} \geq 0$ and $-A'_{xx} \geq 0$. Together with the constraint $A' \succeq 0$, this would imply that $A'_{xx} = A'_{yy} = 0$ and also $A'_{xy} = 0$, but $\nabla^2 G(x', y')$ would still not be positive semidefinite. Since a positive semidefinite Hessian is a necessary condition of minimality in the interior of a feasible region (see for example [2, Prop. 3.3.1]), we conclude that the minimum in (6) cannot occur in the interior of the region $[-1, 1]^2$.

We consider next one edge of the region $[-1, 1]^2$, for example by setting $y' = +1$. This restriction would make $G(x', 1)$ (4) either a concave quadratic function of x' (if $A'_{xx} > 0$) or a linear one ($A'_{xx} = 0$). In either case, the minimum is achieved at an endpoint of the edge, i.e., $(x', y') = (\pm 1, 1)$. It follows that we can restrict attention to the vertices $(\pm 1, \pm 1)$ as claimed.

We then evaluate $G(x', y')$ at the four vertices:

$$\begin{aligned}
&\min_{(x',y')\in[-1,1]^2} G(x', y') \\
&\quad = \min_{(x',y')\in\{\pm 1,\pm 1\}} G(x', y') \\
&\quad = -\frac{1}{2}(A'_{xx} + A'_{yy}) + \min\left\{\epsilon_x\epsilon_y - A'_{xy} \mp \left(\tilde{b}_x + \tilde{b}_y\right), A'_{xy} - \epsilon_x\epsilon_y \mp \left(\tilde{b}_x - \tilde{b}_y\right)\right\} - \tilde{c} \\
&\quad = -\frac{1}{2}(A'_{xx} + A'_{yy}) + \min\left\{\epsilon_x\epsilon_y - A'_{xy} - \left|\tilde{b}_x + \tilde{b}_y\right|, A'_{xy} - \epsilon_x\epsilon_y - \left|\tilde{b}_x - \tilde{b}_y\right|\right\} - \tilde{c}.
\end{aligned}$$

In the second equality above, the first argument of the $\min$ corresponds to x' and y' having the same sign and the second argument to having opposite signs. The last equality is due to $\min\{-x, x\} = -|x|$. □

Elimination of constraint The combination of (6) and (7) yields

$$\begin{aligned}-\tilde{c} &\geq \frac{1}{2}(A'_{xx} + A'_{yy}) - \min\left\{\epsilon_x\epsilon_y - A'_{xy} - \left|\tilde{b}_x + \tilde{b}_y\right|, A'_{xy} - \epsilon_x\epsilon_y - \left|\tilde{b}_x - \tilde{b}_y\right|\right\} \\ &= \frac{1}{2}(A'_{xx} + A'_{yy}) + \max\left\{A'_{xy} - \epsilon_x\epsilon_y + \left|\tilde{b}_x + \tilde{b}_y\right|, \epsilon_x\epsilon_y - A'_{xy} + \left|\tilde{b}_x - \tilde{b}_y\right|\right\}. \end{aligned} \tag{8}$$

Since $-\tilde{c}$ also appears in the objective function (5) being minimized but not in other constraints, (8) must hold with equality. Substituting (8) into (5) yields the simplified problem

$$\min_{A'\succeq 0,\tilde{b}} \frac{1}{3}(A'_{xx} + A'_{yy}) + \max\left\{A'_{xy} - \epsilon_x\epsilon_y + \left|\tilde{b}_x + \tilde{b}_y\right|, \epsilon_x\epsilon_y - A'_{xy} + \left|\tilde{b}_x - \tilde{b}_y\right|\right\}. \tag{9}$$

1.1.3 Solution of simplified problem

First, we observe that the smallest possible contribution from $\tilde{b}$ to the objective in (9) occurs when both $\tilde{b}_x + \tilde{b}_y = 0$ and $\tilde{b}_x - \tilde{b}_y = 0$, implying $\tilde{b}_x = \tilde{b}_y = 0$. Hence

$$\tilde{b}^* = 0 \qquad \Longrightarrow \qquad b'^*_x = m_y\epsilon_x, \quad b'^*_y = m_x\epsilon_y \tag{10}$$

is an optimal solution, further simplifying (9) to

$$\min_{A'\succeq 0} \frac{1}{3}(A'_{xx} + A'_{yy}) + \left|A'_{xy} - \epsilon_x\epsilon_y\right|. \tag{11}$$

Lemma 3. *The optimal solution to* (11) *is* $A'^*_{xx} = A'^*_{yy} = A'^*_{xy} = \epsilon_x\epsilon_y$.

Proof. First, we consider fixing A'_{xy} and optimizing A'_{xx}, A'_{yy} given A'_{xy}. The semidefinite constraint $A' \succeq 0$ is equivalent to the three constraints $A'_{xx} \geq 0$, $A'_{yy} \geq 0$, and $A'_{xx}A'_{yy} \geq A'^2_{xy}$. With A'_{xy} fixed, problem (11) thus reduces to minimizing the sum of A'_{xx} and A'_{yy} subject to a lower bound on their product. The solution is $A'_{xx} = A'_{yy} = \left|A'_{xy}\right|$ (this corresponds to the fact that the rectangle with minimal perimeter for a given area is a square). Substituting this into (11) yields

$$\min_{A'_{xy}} \frac{2}{3}\left|A'_{xy}\right| + \left|A'_{xy} - \epsilon_x\epsilon_y\right|.$$

The solution to this last problem is $A'_{xy} = \epsilon_x\epsilon_y$ (as the only point whose subdifferential $[-1/3, 5/3]$ includes zero). □

In summary, an optimal solution to the problem with transformed variables (x', y') and parameterization in (3) is $A'^*_{xx} = A'^*_{yy} = A'^*_{xy} = \epsilon_x\epsilon_y$, b'^* given by (10), and from (8),

$$\tilde{c}^* = -\epsilon_x\epsilon_y \qquad \Longrightarrow \qquad c'^* = m_xm_y - \epsilon_x\epsilon_y. \tag{12}$$

The optimal mean gap from (5) is

$$\mathbb{E}[G(x', y')] = \frac{2}{3}\epsilon_x\epsilon_y. \tag{13}$$

1.1.4 Transform back to original variables (x, y)

Substituting the optimal parameters in Lemma 3, (10), and (12) into (3) yields

$$\begin{aligned} L(x', y') &= \frac{1}{2}\epsilon_x\epsilon_y \begin{bmatrix} x' & y' \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} m_y\epsilon_x & m_x\epsilon_y \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + m_xm_y - \epsilon_x\epsilon_y \\ &= \frac{1}{2}\epsilon_x\epsilon_y\left(x'^2 + y'^2\right) + \left[\epsilon_x\epsilon_y x'y' + m_y\epsilon_x x' + m_x\epsilon_y + m_xm_y\right] - \epsilon_x\epsilon_y. \end{aligned}$$

Recognizing from (2) that the quantity in square brackets is xy, and inverting (1) to transform the remainder, we obtain (11) in the main text. It can be verified that the mean gap remains $(2/3)\epsilon_x\epsilon_y$.

1.2 Proof of Corollary 1

We make the same change of variables (1) as in the proof of Theorem 1. We then parameterize $U(x', y')$ by the negative of the right-hand side of (3),

$$U(x', y') = -\frac{1}{2}\begin{bmatrix} x' & y' \end{bmatrix}\begin{bmatrix} A'_{xx} & A'_{xy} \\ A'_{xy} & A'_{yy} \end{bmatrix}\begin{bmatrix} x' \\ y' \end{bmatrix} - \begin{bmatrix} b'_x & b'_y \end{bmatrix}\begin{bmatrix} x' \\ y' \end{bmatrix} - c', \qquad A' \succeq 0, \tag{14}$$

where the negation of A and specifically $-A' \preceq 0$ ensure that $U(x', y')$ is concave. The gap $G(x', y') = F(x', y') - L(x', y')$ is then given by the same expression as (4), but with the four terms $\epsilon_x\epsilon_y$, $m_y\epsilon_x$, $m_x\epsilon_y$, $m_x m_y$ all negated. The rest of the proof in Appendix 1.1 goes through with these sign changes, so that the optimal upper bound in the transformed space is

$$\begin{aligned} U(x', y') &= \frac{1}{2}\epsilon_x\epsilon_y\begin{bmatrix} x' & y' \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} m_y\epsilon_x & m_x\epsilon_y \end{bmatrix}\begin{bmatrix} x' \\ y' \end{bmatrix} + m_x m_y + \epsilon_x\epsilon_y \\ &= -\frac{1}{2}\epsilon_x\epsilon_y\left(x'^2 + y'^2\right) + \left[\epsilon_x\epsilon_y x'y' + m_y\epsilon_x x' + m_x\epsilon_y + m_x m_y\right] + \epsilon_x\epsilon_y. \end{aligned}$$

The optimal upper bound in the original space is therefore as shown in (12) in the main text.

2 Experiments

2.1 Computing environment

All verifiers were run using their GPU implementations. The SST experiments were run on a server equipped with an NVIDIA Tesla V100 GPU (32GB). The Yelp experiments were run on a second internal cluster using nodes with A100 GPUs and 80 GB of GPU memory. One CPU and one GPU were used at a time for all experiments. Within each dataset, all experiments were run on the same hardware (in particular, either a V100 or A100 GPU), making for fair comparisons between the verification methods. The code is in the supplementary materials. Models and data can be found at this link[1].

2.2 Setting

In our method, the optimization process for the parameterized bounds is implemented using the built-in Adam [3] optimizer in PyTorch. The stepsize parameter is set to 0.2 during optimization. For PBverifierT, we set the initial value of λ to 0 and the initial value of λ for PBverifierI to -4. The reason for choosing these initial values is explained in Section 6.2 of the main paper, as these values correspond to the minimum mean gap.

2.3 Experimental results

The results of perturbing two positions at a time are presented in Table 1. It can be seen from the table that the results show a similar trend to when only one position is perturbed. Just like when only one position is perturbed, PBverifierI still performs the best.

Table 1: Comparison of certified safe perturbation radii (higher is better) under different settings.

Dataset	N	#	Acc.	Baseline		PBverifierT		PBverifierI	
				Avg	#Win	Avg	#Win	Avg	#Win
Yelp	1	100	91.84	0.2159	38	0.2188	26	0.2163	36
	2	100	91.89	0.1375	0	0.1332	0	0.1426	100
	3	100	92.02	0.0882	0	0.0911	0	0.0942	100
SST	1	75	83.94	0.2486	36	0.2328	3	0.1839	29
	2	82	84.06	0.1622	3	0.1608	0	0.1633	62
	3	82	84.61	0.1344	0	0.1317	0	0.1352	72

[1] https://figshare.com/s/a278da0b75b656985bce

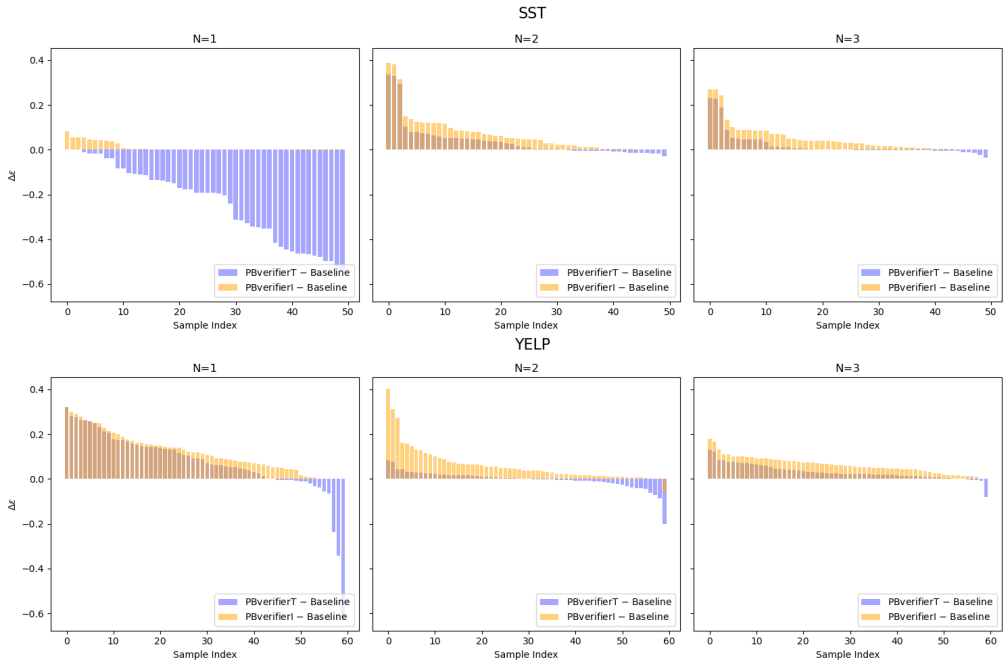


Figure 1: Distribution of maximum robustness radius differences (ℓ_1 perturbation).

Fig 1 and Fig 2 presents the differences of the verification results of our methods and the baseline across all instances under ℓ_1 and ℓ_∞ norm perturbation. The value of each bar represents the difference in the maximum certified radius on the same instance between our two methods and the baseline. The sample indices on the horizontal axis are ordered based on these differences. Points above the difference $= 0$ line indicate that our method can certify a larger perturbation radius for the corresponding instance. As shown in the figure, for most instances, both of our proposed parameterized bounds outperform the baseline. This indicates that flexible parameterized bounds are more advantageous in practical verification tasks.

We used hypothesis testing to investigate whether our method is significantly better than the baseline. We treated the difference in the radius values obtained by the two methods on each instance as a random variable. The null hypothesis was set such that the difference in safe radius is less than or equal to zero. We set the p-value threshold to 0.05, and if the p-value is less than 0.05, it indicates that our method is significantly better than the baseline. Table 2 shows the statistical significance results corresponding to Table 1 in the main paper.

3 Limitation

Our verifier has several limitations which are commonly shared by most existing over-approximation methods. First, The approximation error at each layer of a neural network increases rapidly with the depth of the network, and experimental results suggest that this growth may may follow an exponential trend. Although our method uses parameterized bounds to control the error, for neural networks with more than 10 layers, the verifiable input space still becomes extremely small. Second, due to the fact that the semantics of objects processed by deep learning are difficult to formalize, in most applications we are unable to formally define the complete input space. Instead, we rely on certain inputs with known semantics, along with the input space defined by a ℓ_p ball around them, to formally define a subset of inputs. As a result, most verifiers can only handle input spaces within a certain norm ball.

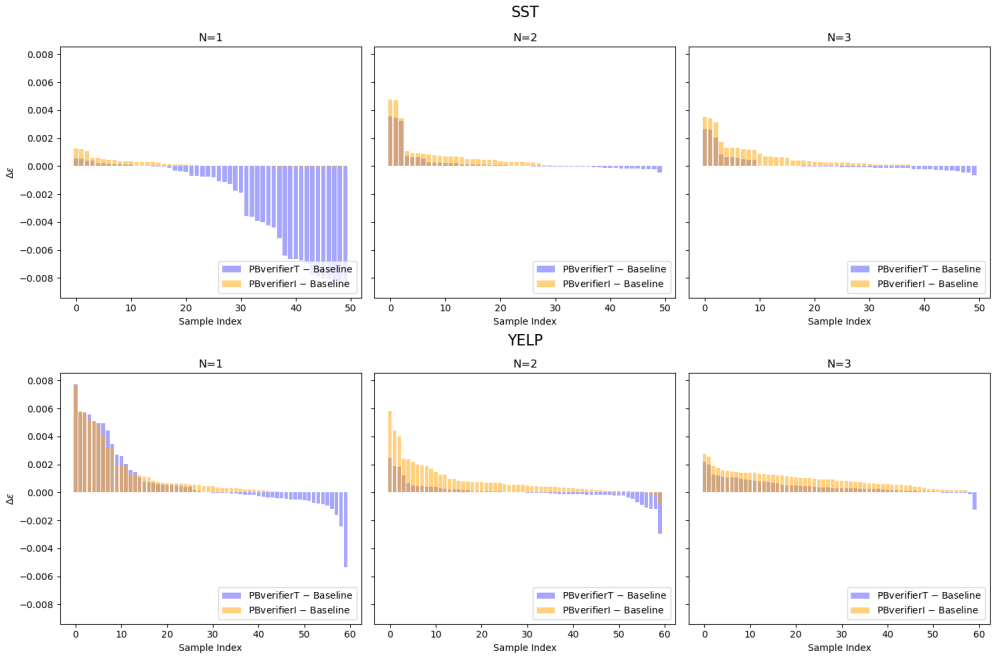


Figure 2: Distribution of maximum robustness radius differences (ℓ_{inf} perturbation).

Table 2: Statistical significance results.

Dataset	N	ℓ_p	Baseline vs. PBverifierT		Baseline vs. PBverifierI	
			Outperform	p-value	Outperform	p-value
Yelp	1	ℓ_1	True	1.8×10^{-4}	True	1.0×10^{-10}
		ℓ_2	True	1.7×10^{-4}	True	3.6×10^{-7}
		ℓ_∞	True	5.6×10^{-3}	True	3.0×10^{-7}
	2	ℓ_1	False	3.7×10^{-1}	True	4.9×10^{-10}
		ℓ_2	False	5.4×10^{-1}	True	4.3×10^{-7}
		ℓ_∞	False	4.2×10^{-1}	True	1.7×10^{-7}
	3	ℓ_1	True	7.0×10^{-10}	True	1.0×10^{-10}
		ℓ_2	True	3.7×10^{-9}	True	1.0×10^{-10}
		ℓ_∞	True	7.6×10^{-9}	True	1.0×10^{-10}
SST	1	ℓ_1	False	9.0×10^{-1}	True	7.5×10^{-5}
		ℓ_2	False	7.5×10^{-1}	True	5.2×10^{-5}
		ℓ_∞	False	9.0×10^{-1}	True	9.0×10^{-5}
	2	ℓ_1	True	6.3×10^{-4}	True	5.9×10^{-7}
		ℓ_2	True	2.9×10^{-3}	True	2.0×10^{-7}
		ℓ_∞	True	2.1×10^{-3}	True	2.2×10^{-5}
	3	ℓ_1	True	7.3×10^{-4}	True	7.7×10^{-7}
		ℓ_2	False	1.2×10^{-1}	True	1.2×10^{-6}
		ℓ_∞	True	1.1×10^{-1}	True	7.2×10^{-7}

4 Broader Impact

In this paper, we propose a parameterized and adjustable bounding scheme for the attention mechanism, the core component of transformer architectures. This technique enables rigorous analysis of the output range of each network layer under input perturbations constrained within an ℓ_p ball. It can be used in a wide variety of tasks ranging from robustness verification to certified defense, and potentially many more applications requiring a provable perturbation analysis. It is promising for several safety-critical ML applications, such as transportation, engineering, and healthcare, etc. We

expect that our framework will significantly improve the robustness and reliability of real-world ML systems with theoretical guarantees.

References

[1] Stephen P Boyd and Lieven Vandenberghe. *Convex optimization*. Cambridge University Press, 2004.

[2] Dimitri P. Bertsekas. *Nonlinear Programming*. Athena Scientific, Belmont, MA, USA, 2nd edition, 1999.

[3] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In Yoshua Bengio and Yann LeCun, editors, *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015.